

Alvaro Guerra

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EDUCATION

Texas A&M University

B.S. Mechanical Engineering

College Station, TX

2021 - 2025

EXPERIENCE

Structures Lead

IGNITORS (Rocketry Team)

11-2024 – 06-2025

College Station, TX

- Delivered presentations on composite manufacturing, structural aspects of rocketry, and IREC competition strategies, aligning team tasks with project timelines to ensure efficient and informed progress toward mission goals
- Designed and fabricated composite-based rocket to a planned altitude of 11,000 feet

Structures and Material Science Designer

SAE Aero Design Team

05-2022 – 04-2023

College Station, TX

- Secured 1st Place Finalist position at the SAE Aero West International Competition through aircraft design, topological optimization, and teamwork.
- Optimized airplane payload, 58% of total weight, using SolidWorks and ANSYS, adjusting the center of gravity to be 9.5 inches aft of the main wing leading edge for CFD analysis

Garden Robot Developer

Garden Cultivation Capstone Project

08-2024 – 05-2025

College Station, TX

- RC robot empowering individuals with physical disabilities to garden independently—met 100 % of design specs and came in \$200 under budget
- Implemented Arduino code for robotic wheel movement, watering, fertilizing, and climate sensing that improved consumable volume efficiency by 26%

Real Estate Analyst Intern

River Valley Appraisals of Texas, LLC

05-2021 – 08-2021

Edinburg, TX

- Conducted residential property evaluations using regression analysis and comparable sales market data with excel data analysis

PROJECTS

openLayup (Python)

April 2025

- Developing an open-source structural simulation tool for rocket components, leveraging Python and APIs (Onshape, SolidWorks, Granta Ansys) for material property retrieval, geometry integration, and failure analysis
- Implemented failure mode analysis using Max Stress/Strain criteria, CLT, Tsai-Wu, Tsai-Hill, Hashin-Rotem, and Puck Criterion

BeamMeUpScotty (Python, Jupyter)

April 2025

- 2D structural analysis tool that enables users to model, analyze, and visualize trusses and beams
- Implemented numerical methods for analytical estimation using Euler-Bernoulli, stiffness matrices, and Newton-Raphson techniques

TECHNICAL SKILLS

Software: SolidWorks, Onshape, ANSYS, Python, Matlab, HTML/CSS, Git, Microsoft Office

Manufacturing: 3D Printing (Prusa/Ender3), Hand Tools, CNC Machining, Manual Mill/Lathe

Libraries: pandas, NumPy, Matplotlib, BeautifulSoup, PyQt